

Referee report on IER-31540-1-manuscript “NIMBYism and the Housing Supply”

Summary

The paper develops a political economy model of the housing market in which the supply of new homes is determined by a median vote among households. Voter support for additional housing is determined by the effect this vote has on households' expected discounted utility via changes in house prices. Households that own housing, and therefore typically experience financial losses from additional supply, generally oppose new housing. The demographic structure of the population thus should have effects on house prices. The authors explore this hypothesis by introducing a model of median voter to an OLG model of house prices and use the model to estimate the size of this effect on house prices in the US since 1950 and provide projections into the future.

Main comments and assessment

I find the question of the paper very interesting and was excited to read the paper. However, my excitement was not met by the current version of the paper. The current draft leaves out many details of the model, calibration and the quantitative findings that I think are required to make the reader understand and value their exercise. The draft is also somewhat sloppy in places (some of my Minor comments).

The authors need to provide more detailed description of calibration and explain the details of the quantitative exercise. The least satisfactory part, to me, is the steady-state analysis. The authors need to be explicit about what they mean by that and why a steady-state analysis is the relevant exercise given their question which deals with demographic developments since 1950.

This brings me to my main comment, which has a bearing on both the steady-state analysis as well as the other analysis the authors carry out: nowhere in the paper is a description of the demographic developments in the model, despite the paper being about demographic changes affecting house prices. What is the law of motion of the shares of people of different ages in the population? This law of motion should be also related to survival probabilities, which also seem to be missing. Next, how do people forecast the future evolution of demographics and thus future prices (house prices are assumed to be forecast as random walk---is this reasonable?---but what about rents and interest rates)?

These details need to be provided in future revisions.

Minor comments

- Fig 1: What data on rents for non-US countries do you use? Rents data going as far as in the sample are typically unreliable. In any case, there is no need to include Figure 1 in the manuscript anyway and you should drop it. The motivation is compelling even without that Figure.
- On p. 4 (the last para), I suggest you start "We use the model to..." instead of "In this paper..."
- On p. 5 at the bottom "The difficulty in providing" It is not clear how this is related to what you said earlier: "However, establishing a direct link....". These two statements should be more clearly related to each other.
- In the literature review section, the two paragraphs starting with "Housing politics is a good candidate...." and "Numerous surveys..." are only broadly related to the paper and make the section unnecessarily long. I suggest you drop them.
- Model and calibration
 - In the lifetime utility there are no survival probabilities---baby boomers are characterized not only by coming from a high-fertility period but also by enjoying advances in longevity. That makes them stay around longer and also affects their saving behavior. You should either incorporate survival probabilities into the model or have a strong argument why they are inconsequential for the results.
 - I did not get why the omega upper bar is necessary in the bequest function; would not decreasing marginal utility be present even without that parameter?
 - If income is stochastic, why is there no expectations operator in from of the lifetime utility function?
 - At the point of describing the income process on p.8, either do not go into any details yet or elaborate further on what you mean by eg "lowest income states"
 - You should explain why you need the transaction costs in the housing market
 - It is more correct to say that the households can purchase shares in the real estate firm, rather than the firm issues a debt, as the rate of return in equilibrium depends on state variables—it means the deposits of the households in the real estate firm do not carry any promised interest rate, as would be the case with debt

- You should explain why you need the cost of providing rental services in equation (13)
- You should specify in Section 3.2 whether the real estate firms compete with owner-occupiers for the aggregate housing stock or whether there is a separate stock of units that are just for rental—in other words, if the housing stock market is integrated or segmented
- You need to provide details of how you model the provision of new housing (“This is similar to Kaplan et al. (2020), who” is not sufficient)
- I did not get this statement on p.11: “An increase in the housing supply will also affect the price of rental properties by lowering the input price for the real estate firms.”
- First sentence in Section 3.5, there seems to be “denote” missing in the sentence.
- In Section 4 you need to provide details of your income process, especially in relation to what you said earlier.
- Steady state: a law of motion of the age distribution in the population generally implies a non-zero population growth rate. For the stationary distribution, it is one of the eigenvalues of the transition matrix. But then everything in steady state needs to be normalized by the growth rate to achieve stationarity of the model. Which also means that there needs to be a steady state housing stock growth that somehow should keep housing per capita constant. It is not clear how any of this is treated in the model, which then leaves the reader wondering what exactly are the authors doing and how the model works.